

Product Requirements Document

DASHBOARD - Display

Issued by	Role/Department	Date
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REVISION TABLE

Version	Date	Issued by	Description
A-1	09/06/2025	Alessandro Zingaretti	Update to the 2025 configuration
A-2			
A-3			
B-1			
B-2			
B-3			
B-4			

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GENERAL PRODUCT DESCRIPTION

Product Description	
Key Features/ Functionality	Showing car parameters on the steering wheel display (car speed, critical temperatures, R2D, SOC and power consumption,...)
HV/LV	LV
Software	Nextion Editor, STM32CubeIDE
Software Language	C/C++, Nextion language
Other components Compatibility, Ecosystem, etc.	74HC165 (shift register), MCP2551 (CAN-Bus transceiver)
Installation (where)	NX4832T035, STM32F303K8T6
Dimensions and Tolerances	Nextion: 100.5 (L)×54.94 (W)×5.45 (H) (.X±,20) Stm32:
Connections	Power, GND, CAN.
Weight	Nextion: 48.2g

REGULATORY

Regulatory Requirements	
Safety	
Structural	
Electrical	
Labeling	
Regulatory Marks	
Place of Origin	
Serial Number	
Financials	
BOM Cost	
COGs	
MSRP	
Volume	
Number of pieces	2
Spare Parts	1

INPUT AND OUTPUT

Product Description	
Data Input (CAN/Bluetooth....)	Stm: CAN-BUS Nextion: Serial (USART)
Signal Input	Stm: GPIO (<u>Shift register</u> , buttons)
Data Output (CAN/Bluetooth....)	Stm: CAN-bus, Serial (USART)

Signal Output	
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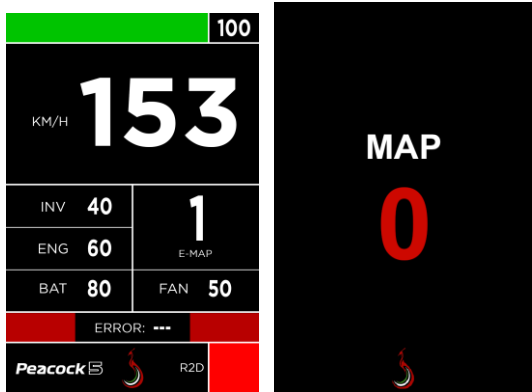
ENVIRONMENT

Operating Environment	
Where in the car	Steering Wheel
Temperature Range	Nextion: (-20,+70)°C Stm: (-40,+85)°C
EMI (vulnerability to electromagnetic interference)	

INTERFACE

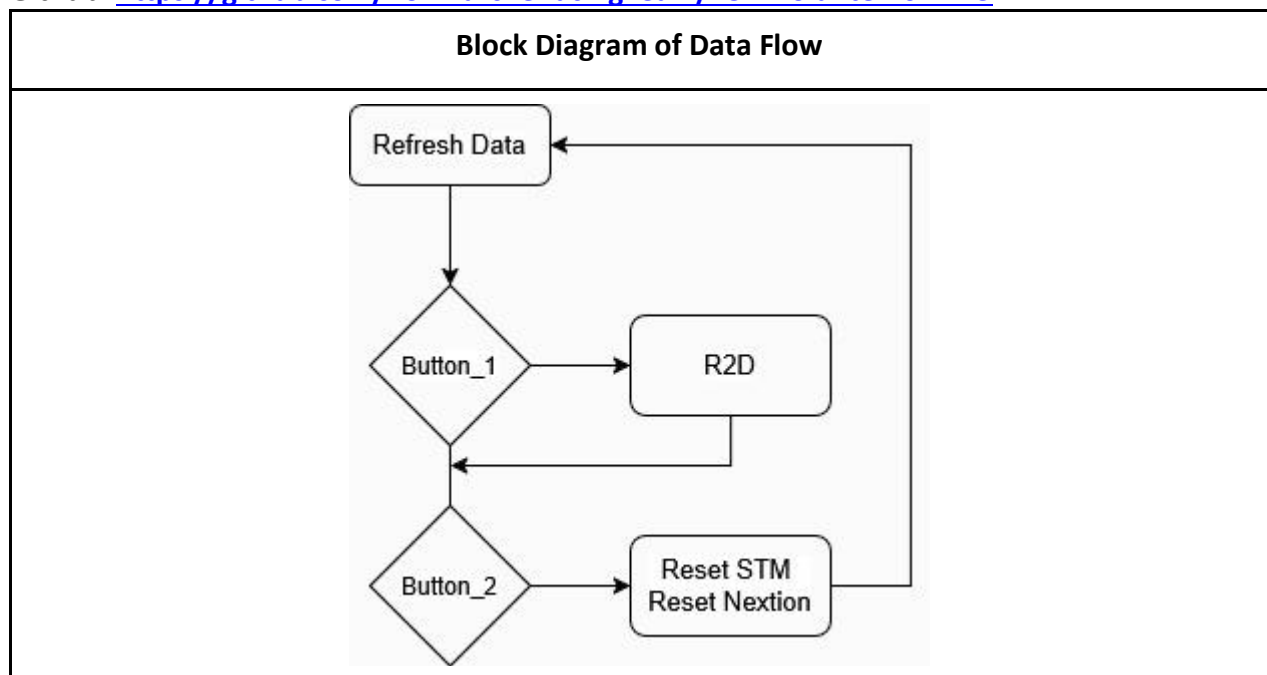
User Interface	
Communication protocol	
Communication Input	
Communication output	
Connector	
Visual Interface (Screen size and type, LEDs)	Nextion: 100.5 (L)×54.94 (W)×5.45 (H) (.X±,20)
External Software interface	STM32CubeIDE NEXTION EDITOR

DESIGN

Renderings	
Color, Material, and Finish (CMF)	
Logo size and placement	

SOFTWARE ARCHITECTURE AND DATA PROCESSING

Github: <https://github.com/PolimarcheRacingTeam/LCD-Volante-2024-25>



The code is used to manage the communication between the car and the user interface on the steering wheel. (display and buttons)

The software receive from the car's CAN-bus the data to be shown on the display. The data to be displayed are: Speed, SoC, inverter and motor temperature....

Once received and unpacked, they must be sent via serial to the display. In addition, the map selected by the hand controller is displayed.

The value of the hand controller is managed by a shift register (see above for the datasheet) and has seven possible positions. Each position corresponds to a map for the engine control, which the driver can choose by turning the knob. The code reads the value of the selected map and send it both via CAN (0x040) and via serial for the display. In particular:

Hand controller's Position	Display message	CAN message
1	Map 1 (0%)	0
2	Map 2 (11-30%)	1
3	Map 3 (30%-60%)	2
4	Map 4 (60%-100%)	3
5	Error: Map not available!	7
6	Error: Map not available!	7
7	Error: Map not available!	7

Button 1

This button manages the R2D (EV regulation 4.11). When the car reads the button pressed and the brake message (0x012) high, the R2D parameter (0x016) is set high.

Brakes (0x012)	Button 1	R2D (0x016)
0	0	0
0	1	0
1	0	0
1	1	1

Button 2

This button manages a software reset for the nextion display and then for the STM32

ERRORS

Errors Parameters Definition	
Critical parameter	The script must be able to show the numerical value of any error if sent via CANbus

ELECTRICAL HARDWARE AND SENSORS

Block Diagram of Electrical Hardware	
Supply Voltage	
Supply Current	
Output	
OverVoltage/ OverCurrent / Short Circuit Protection on power supply	

PRODUCT VALIDATION / QUALIFICATION

Data Test	
Visualization Test	
Ergonomics Test	
Utility Test	